

Course Title: Linear Algebra and Transform Techniques

MCQ

UNIT I	
1.	Rank of an identity matrix of order 4 is _____ a) 4 b) 3 c) 1 d) 0
2.	System of equations $AX=B$ has solutions if a) $\text{Rank}[A] = \text{number of unknowns}$ c) $\text{Rank}[A] \neq \text{number of unknowns}$ b) $\text{Rank}[A] = \text{Rank}[A \mid B] \leq \text{number of unknowns}$ d) $\text{Rank}[A] \neq \text{Rank}[A \mid B]$
3.	System $AX=0$ has nontrivial solution if and only if a) Determinant of A = Determinant of X b) Determinant of $A = 0$ c) $\text{Rank}[A] \neq \text{Rank}[A \mid B]$ d) Determinant of $A \neq 0$
4.	If $[A:B] = \begin{bmatrix} 2 & 3 & -1 & : & 5 \\ 0 & 1 & 3 & : & -1 \\ 0 & 0 & \frac{k}{3} & - & 6 & : & 5 \end{bmatrix}$ then the value of k for which the system is inconsistent is _____ a) 18 b) -18 c) 6 d) -6
5.	System of equations $AX=B$ has infinite number of solutions if a) $\text{Rank}[A] = \text{Rank}[A \mid B] = \text{number of unknowns}$ b) $\text{Rank}[A] = \text{Rank}[A \mid B] < \text{number of unknowns}$ c) $\text{Rank}[A] \neq \text{Rank}[A \mid B]$ d) $\text{Rank}[A] = \text{Rank}[A \mid B] > \text{number of unknowns}$
6.	In the LU factorization of $A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$, the matrix U is a) $U = \begin{bmatrix} 3 & 7 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ b) $U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & 2 \\ 0 & 0 & -1 \end{bmatrix}$ c) $U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ d) $U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
7.	The eigen values of $A = \begin{bmatrix} 5 & 6 & 17 \\ 0 & -19 & 23 \\ 0 & 0 & 37 \end{bmatrix}$ are a) -19, 37, 5 b) 2, -3, 7 c) 5, 19, 37 d) 0, 5, 17
8.	Eigen value of a triangular matrix A are a) diagonal element of A b) sum of diagonal element of A c) Determinant of A d) Trace of A
9.	Eigenvectors corresponding to distinct eigenvalues are:

	a) Always parallel	b) Always orthogonal
	c) Linearly independent	d) Linearly dependent
10.	Let $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$, the eigen value $\lambda_1 = 2, \lambda_2 = 3$ and $\lambda_3 = \underline{\hspace{2cm}}$	
	a) 3	b) 2
	c) 5	d) 7
11.	If A is a square matrix, then $\det(A) =$	
	a) sum of the eigen values of A	b) product of eigen values of A
	c) product of diagonal elements of A	d) sum of diagonal elements of A
12.	For which value of k the system of two equations $x + y = 1$ $2x + 2y = k$ has infinite solutions:	
	a) $k = 0$	b) $k = 1$
	c) $k = 2$	d) any value of $k \neq 1$
13.	If A is 3×4 matrix the maximum possible value of the rank A is	
	a) 4	b) 3
	c) 2	d) 0
14.	If a matrix A is reduced to row echelon form, then rank of the matrix is	
	a) number of zero rows	b) number of non-zero rows
	c) total number of rows	d) total number of columns
15.	The characteristic equation of the matrix $A = \begin{bmatrix} 2 & 0 \\ 4 & 7 \end{bmatrix}$ is:	
	a) $\lambda^2 - 9\lambda + 14 = 0$	b) $\lambda^2 - 14\lambda + 9 = 0$
	c) $\lambda^2 - 9\lambda - 14 = 0$	d) $\lambda^2 - 14\lambda - 9 = 0$
16.	If a matrix has distinct eigenvalues, then its minimal polynomial is	
	a) Product of repeated factors	b) Product of distinct linear factors
	c) Constant	d) Quadratic only
17.	Eigen values of $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ are	
	a) 3, -3	b) -3, -5
	c) 3, 5	d) 5, 0
18.	The minimal polynomial of $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$ is	
	a) $m(t) = (\lambda + 2)(\lambda + 9)$	b) $m(t) = (\lambda - 2)(\lambda - 9)$
	c) $m(t) = (\lambda - 2)(\lambda + 9)$	d) $m(t) = (\lambda - 3)(\lambda - 8)$
19.	Which of the following is not a valid elementary operation?	
	a) $R_i \leftrightarrow R_j$	b) $R_i \rightarrow R_j + kR_i$
	c) $R_i \rightarrow kR_i$	d) $R_i \rightarrow 1 + kR_i$
20.	The value of h for which the augmented matrix $\begin{bmatrix} 1 & -3 & -2 \\ 5 & h & -7 \end{bmatrix}$ of a linear system is inconsistent	

	a) $h = -15$	b) $h \neq -15$
	c) $h = 5$	d) $h = -1$
21.	The characteristic equation of a matrix A is obtained from:	
	a) $ A = 0$	b) $ A - \lambda I = 0$
	c) $A - \lambda I = 0$	d) $ A + \lambda I = 0$
22.	Rayleigh's power method is used to find the	
	a) all eigenvalues of a matrix	b) smallest eigenvalue of a matrix
	c) largest eigenvalue of a matrix	d) complex eigenvalues of a matrix

	UNIT II- VECTOR SPACES	
1.	A subset S of vector space V over field F is a basis of V if S is linearly independent and	
	a)) $Span(S) = F$	b) $Span(V) = S$
	c) $Span(S) = V$	d) $Span(S) = S$
2.	If $\{u_1, u_2, \dots, u_n\}$ is linearly independent set, then $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0$ has	
	a) infinitely many solutions	b) unique nonzero solution
	c) only the trivial solution	d) at least one non-zero solution
3.	Let A be an $m \times n$ real matrix. The row space of A is a subspace of	
	a) R^n	b) the set of all $m \times n$ matrices
	c) R^m	d) R
4.	In two-dimensional Euclidean space $\mathbb{R}^2$, the identity element with respect to vector addition is	
	a) (1,1)	b) (0,0)
	c) (-1,-1)	d) does not exist
5.	In three-dimensional Euclidean space $\mathbb{R}^3$, inverse of $(-2, -2, 0)$ with respect to vector addition is	
	a) (0,0,0)	b) (2,2,2)
	c) (2,2, -2)	d) (2,2,0)
6.	Which of the following sets is linearly dependent?	
	a) $\{(0,0,0), (3,0,3)\}$	b) $\{(1,1,0), (0,0,1)\}$
	c) $\{(0,0,2), (3,0,0)\}$	d) $\{(1,0,1), (0,1,1)\}$
7.	The dimension of the subspace of $\mathbb{R}^3$, spanned by the vectors $(2,0,1), (5,2,1), (1,2, -1)$ is :	
	a) 3	b) 2
	c) 1	d) 0
8.	Let A be an $m \times n$ real matrix. The column space of A is a subspace of	
	a) R^m	b) R^n

	c) R^{mn}	d) R^{m+n}
9.	The set $W = \{a, 0, 0\} : a \in R\}$ is a subspace of:	
	a) R^2	b) R
	c) R^3	d) R^4
10.	If W is a subspace of vector space V then	
	a) $\dim(W) \geq \dim(V)$	b) $\dim(W) \leq \dim(V)$
	c) $\dim(W) \neq \dim(V)$	d) $\dim(W) > \dim(V)$
11.	If a basis of a vector space contains exactly three vectors, then the dimension of the vector space is:	
	a) less than 3	b) greater than 3
	c) 3	d) 0
12.	Dimension of the Null space of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$ is	
	a) 0	b) 1
	c) 2	d) 3
13.	The set $H = \{(a, a) / a \in R\}$, which of the following is true:	
	a) H is a subset of R^2 , but not subspace.	b) H is an empty set
	c) H is a subspace of R	d) H is a subspace of R^2
14.	The set of all solution to the homogeneous equation system $AX = 0$, where A be an $m \times n$ matrix is	
	a) Null space of A	b) Column space of A
	c) Row space of A	d) Null space of A and column space of A
15.	Which of the following is a linear combination of $u = (1, 1, 0)$ and $v = (0, 0, 1)$?	
	a) $(2, 2, 3)$	b) $(2, 0, 1)$
	c) both (a) & (b)	d) neither (a) nor (b)
16.	If $S = \{v_1, v_2 \dots v_n\}$ is a set of vectors in a finite dimension vector space V , then S is called a basis for V if	

	a) S spans V	b) S is linearly independent
	c) either (a) or (b)	d) both (a) and (b)
17.	If a vector can be expressed as a linear combination of other vectors in a set, then the set is	
	a) Linearly independent	b) Orthonormal
	c) Linearly dependent	d) A basis
18.	A set of vectors which is both linearly independent and spans the vector space is called	
	a) Spanning set	b) Coordinate system
	c) Basis	d) Subspace
19.	Which of the following sets forms a basis for the column space of a matrix A ?	
	a) All columns of A	b) All non-zero columns of A
	c) The pivot columns of A	d) The rows of A
20.	The vectors $(1,0,1)$, $(0,1,0)$, $(2,3,2)$ are	
	a) Linearly independent	b) Linearly dependent
	c) Form a basis of $\mathbb{R}^4$	d) Cannot be determined

LINEAR TRANSFORMATIONS

1. Let $T: R^4 \rightarrow R^4$ be the linear transformation defined by $T(x_1, x_2, x_3, x_4) = (0, x_1, x_2, x_3)$.

Then null space of T consists of vectors of the form

- a) $(0,0,0,x_4)$ b) $(x, 0,0,0)$ where x is a real number
c) $(x_1, x_2, x_3, 0)$ d) $(x_1, x_1, x_3, 0)$

2. Let $T: R^3 \rightarrow R^4$ be the linear transformation defined by $T(x_1, x_2, x_3) = (x_1, 0, x_2, x_2)$.

Then basis for null space of T is

- a) $(1,0,0)$ b) $(0,0,1)$ c) $(0,1,0,0)$ d) $(1,1,0,0)$

3. Which of the following statement is **not true** if $T: U \rightarrow V$ is a linear transformation.

a) $T(x.y) = T(x).T(y) \quad \forall x, y \in U$

b) $T(-x) = -T(x) \quad \forall x \in U$

c) $T(0) = 0$ for $0 \in U$

d) $T(2x) = 2T(x) \quad \forall x \in U$

4. Which of the following mappings is **not** a linear transformation?

a) $T(x, y) = (x + y, x - y)$

b) $T(x, y) = (2x, 3y)$

c) $T(x, y) = (x^2, y)$

d) $T(x, y) = (0, x)$

5. Which of the following statement is true if $T: U \rightarrow V$ is a linear transformation

a) Range space of T is a subspace of V

b) Null space of T is a subspace of V

c) V is always the range of T

d) Range space of T is a subspace of U

6. Which of the following statement is true if $T: U \rightarrow V$ is a linear transformation

a) Range space of T is a subspace of U

b) Null space of T is a subspace of U

c) V is always the range of T

d) Null space of T is a subspace of V

7. If $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$ then inner product of u and v is

a) 9

b) 19

c) -3

d) 3

8. Let $T: R^2 \rightarrow R^2$ be the linear transformation defined by $T(x, y) = (x, 0)$. Then matrix of linear transformation with respect to standard basis is

a) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 0 & -1 \\ 0 & -1 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

d) $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$

9. Let $T(x, y) = (x + y, x - y, y)$ be the linear transformation. Then nullity of T is

a) 0

b) 1

c) 2

d) 3

10. Let $T: R^3 \rightarrow R^3$ be the linear transformation with nullity of T is 2 then rank of T is

a) 0

b) 1

c) 2

d) 3

11. Let $T(x, y) = (x + y, x - y, y)$ be the linear transformation. Then null space of T is

a) $\{(x, 0)\}$

b) $\{(0, y)\}$

c) $\{(0, 0)\}$

d) $\{(x, y)\}$

12. The null space of the linear transformation $T(x, y, z) = (x + y, x - y, x + 2z)$ is

- a) $\{(x, 0, z)\}$ b) $\{(0, y, z)\}$ c) $\{(0, 0, 1)\}$ d) $\{(0, 0, 0)\}$

13. The range space of the linear transformation $T(x, y, z) = (x + y, x - y, x + 2z)$ is

- a) $\{(x + y, 0, 2z)\}$ b) $\{(x, y, z)\}$
c) $\{(x + y, x - y, x + 2z)\}$ d) $\{(y, -y, x)\}$

14. If $T: R^4 \rightarrow R^4$ is a linear transformation with Nullity 2, then dimension of range space of T is

- a) 4 b) 2 c) 1 d) none of these

15. Range space of linear transformation $T: U \rightarrow V$ is a subspace of

- a) V b) U c) U and V d) none of these

16. Which of the following set of vectors is not orthogonal

- a) $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ b) $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ c) $\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ d) $\left\{ \begin{bmatrix} -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

17. If $T: R^6 \rightarrow R^6$ is a linear transformation with Rank of T is 2, nullity of T is

- a) 4 b) 2 c) 1 d) none of these

18. Which of the following statement is true if $T: V \rightarrow W$ is a linear transformation

- a) Range space of T is a subspace of V
b) Null space of T is a subspace of W

c) Range of T is a subset of W

d) cT is not a linear transformation from V to W , where c is any scalar.

19. Let $T: R^6 \rightarrow R^6$ be the linear transformation with nullity of T is 2 then rank of T is

a) 4

b) 1

c) 2

d) 3

20. Two vectors u and v in R^n are orthogonal if

a) $u \cdot v = v \cdot u$

b) $u \cdot v = 1$

c) $u \cdot v \geq 0$

d) $u \cdot v = 0$

MAT106-UNIT IV- MULTIPLE CHOICE QUESTIONS

1.	If a square matrix A is diagonalizable, then for any positive integer $k > 2$:	
1	A) $A^k = PD^kP^{-1}$	B) $A^k = P^kD P^{-k}$
2.	C) $A^k = P^kD^kP^{-k}$	D) $A^k = P^kD^k$
3.	Let A be an $n \times n$ matrix, then A is diagonalizable if and only if	
	A) A is zero matrix	B) A has linearly independent columns
	C) A has n linearly independent eigenvectors	D) A has only $(n - 1)$ linearly independent vectors.
4.	If an $n \times n$ matrix A has n distinct eigen values, then A is:	
	A) symmetric matrix	B) diagonalizable matrix
	C) not diagonalizable matrix	D) skew symmetric matrix
5.	Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, if $A = PDP^{-1}$ with $P = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, then D is:	
	A) $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$
	C) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
6.	The quadratic form $3x_1^2 + 5x_2^2 + 6x_3^2$ is:	
	A) Positive definite	B) Negative definite
	C) Indefinite	D) Negative semi-definite
7.	Matrix of the quadratic form $2x_1^2 - 4x_1x_2 + x_2^2$ is:	
	A) $\begin{bmatrix} 2 & -4 \\ -4 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 2 & -2 \\ -2 & 1 \end{bmatrix}$
	C) $\begin{bmatrix} 1 & -4 \\ -4 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 2 & -2 \\ -4 & 2 \end{bmatrix}$
8.	If eigen values of the quadratic form $Q(X)$ are $-3, -2$ and 0 , then the quadratic form $Q(X)$ is:	
	A) Positive definite	B) Negative definite
	C) Indefinite	D) Negative semi-definite
9.	Eigen values of the quadratic form $x_1^2 + 2x_1x_2 + x_2^2$ are	
	A) $0, 2$	B) $1, 1$
	C) $1, 2$	D) $1, 0$
10.	Matrix of the quadratic form $x_1^2 - 2x_2^2 + 3x_3^2 - 2x_1x_3$ is	
	A) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$	B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$
	C) $\begin{bmatrix} 1 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & 3 \end{bmatrix}$	D) $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$
11.	In QR factorization of matrix A , Q is a matrix whose columns form an orthonormal basis for	
	A) Column space of A	B) Null space of A
	C) Row space of A	D) Both null space and row space of A
12.	The quadratic form $3x_1^2 - 4x_1x_2 + 6x_2^2$ is	
	A) Positive definite	B) Positive semi-definite
	C) Negative definite	D) Indefinite
13.	In QR factorization of matrix A , R is equal to:	
	A) $Q^T A^T$	B) $A^T Q$
	C) QA	D) $Q^T A$
14.	If $A = QR$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, then $R =$	
	A) $\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$	B) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
	C) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
15.	The QR factorization of an $m \times n$ matrix A exists, if	
	A) the columns of A are linearly independent	B) the rows of A are dependent

	C) A has non-zero columns	D) rank of A is $n - 1$
16.	The set of least-squares solutions of $Ax = b$ coincides with the nonempty set of solutions of the normal equation:	
	A) $AA^T x = A^T b$	B) $A^T Ax = A^T b$
	C) $A^T x = A^T b$	D) $A^T x = b$
17.	If $\hat{x}$ is the least square solution of the inconsistent system $Ax = b$, then	
	A) $b - A\hat{x}$ is orthogonal to rows of A	B) $b - A\hat{x}$ is in the null space of A
	C) $b - A\hat{x}$ is in the row space of A	D) $b - A\hat{x}$ is orthogonal to columns of A
18.	A matrix $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ has the singular value decomposition $A = U\Sigma V^T$ with the singular values $\sqrt{5}$ and $\sqrt{45}$ then $\Sigma =$	
	A) $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$	B) $\begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$
	C) $\begin{bmatrix} \sqrt{5} & 0 \\ 0 & 5 \end{bmatrix}$	D) $\begin{bmatrix} \sqrt{35} & 0 \\ 0 & 5 \end{bmatrix}$
19.	Singular values of $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ are	
	A) $1, -3$	B) $9, 1$
	C) $\sqrt{3}, 1$	D) $3, 1$
20.	Singular value of matrix A are	
	A) the eigen values of the matrix $A^T A$	B) the eigenvalues of A
	C) the positive square roots of the eigenvalues of $A^T A$	D) the square roots of the eigen values of A
21.	If a matrix A is diagonalizable, then the Jordan Canonical Form of A is	
	A) an upper triangular matrix	B) a lower triangular matrix
	C) a diagonal matrix	D) a matrix with only 1s on the super diagonal

MAT106 - Fourier Series and Integral
(II sem - Common to IT Branches)

1.	If $F\{f(t)\} = F(\omega)$, then $F\{f(at)\} =$ _____			
	A)	$F\left(\frac{\omega}{a}\right)$	B)	$a F(a\omega)$
	C)	$\frac{1}{a} F\left(\frac{\omega}{a}\right)$	D)	$F(a\omega)$
2.	The Fourier series expansion of x^3 in $(-1, 1)$ with $f(x+2) = f(x)$ has			
	A)	Only sine terms	B)	Only cosine terms
	C)	Both sine and cosine terms	D)	Only sine terms and a non-zero constant
3.	The value of a_0 in the Fourier series of the function $f(x) = x$ in the interval $(-1, 1)$ is _____			
	A)	$\sum_{n=0}^{\infty} \frac{2}{n^2} \cos \frac{n\pi x}{2}$	B)	$\sum_{n=0}^{\infty} \frac{2}{n} \sin \frac{n\pi x}{2}$
	C)	0	D)	1
4.	If $f(x)$ is a periodic function in $(-2, 2)$, then $b_n =$ _____			
	A)	$\frac{1}{2} \int_{-2}^2 f(x) dx$	B)	$\frac{1}{2} \int_{-2}^2 f(x) \cos nx dx$
	C)	$\frac{1}{2} \int_{-2}^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx$	D)	$\int_{-2}^2 f(x) \sin nx dx$
5.	If $f(x)$ is a periodic function in $(-\pi, \pi)$, then $a_0 =$ _____			
	A)	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$	B)	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
	C)	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$	D)	$\int_{-\pi}^{\pi} f(x) \cos nx dx$
6.	Which of the following function is odd			
	A)	$f(x) = \begin{cases} 1+x, & -\pi < x < 0 \\ 1-x, & 0 < x < \pi \end{cases}$	B)	$f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$
	C)	$f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$	D)	$f(x) = \begin{cases} -2x, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$
7.	If $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$ then its Fourier transform contains			
	A)	only sine terms	B)	only cosine terms
	C)	both sine and cosine terms	D)	neither sine nor cosine terms
8.	If $f(x)$ is an even function, the Fourier transform integral reduces to			
	A)	$\int_0^{\infty} f(x) \sin(ax) dx$	B)	$\int_{-\infty}^{\infty} f(x) e^{-iax} dx$
	C)	$2 \int_0^{\infty} f(x) \cos(ax) dx$	D)	$2 \int_{-\infty}^0 f(x) \cos(ax) dx$

9.	If $f(x)$ is even, which integral is zero?			
	A)	$\int_{-\infty}^{\infty} f(x) \sin(\alpha x) dx$	B)	$\int_{-\infty}^{\infty} f(x) \cos(\alpha x) dx$
	C)	$\int_0^{\infty} f(x) \cos(\alpha x) dx$	D)	$\int_0^{\infty} f(x) \sin(\alpha x) dx$
10.	$L\left\{\sin\left(\frac{\pi}{4}\right)\right\}$ is -----			
	A)	$\frac{\sqrt{2}}{s}$	B)	$\frac{1}{\sqrt{2}s}$
	C)	$\frac{1}{s}$	D)	$\sqrt{2}s$
11.	$L\{5^t\}$ is -----			
	A)	$\frac{1}{5s}$	B)	$\frac{1}{s - \log 5}$
	C)	$\frac{1}{s-5}$	D)	$\frac{1}{s^2-5}$
12.	If $L\{f(t)\} = \bar{f}(s)$, then $L\{tf(t)\} =$ -----			
	A)	$s\bar{f}(s)$	B)	$\frac{d}{ds}(\bar{f}(s))$
	C)	$(-1)^n \frac{d}{ds^n}(\bar{f}(s))$	D)	$(-1)^n \frac{d^n}{ds^n}(\bar{f}(s))$
13.	If $L\{f(t)\} = \bar{f}(s)$, then $L\left\{\frac{f(t)}{t}\right\} =$ -----			
	A)	$\frac{1}{s}\bar{f}(s)$	B)	$\int_s^{\infty} \bar{f}(s) ds$
	C)	$\frac{1}{s} \int_s^{\infty} \bar{f}(s) ds$	D)	$\int_0^{\infty} \bar{f}(s) ds$
14.	$L\{10t\} =$ -----			
	A)	$\frac{10}{s}$	B)	$\frac{10}{s^2}$
	C)	$\frac{20}{s^2}$	D)	$10s$
15.	If $L\{f(t)\} = \bar{f}(s)$, then $L\left\{\int_0^t f(t) dt\right\} =$ -----			
	A)	$\int_s^{\infty} \bar{f}(s) ds$	B)	$s\bar{f}(s)$
	C)	$\int_0^{\infty} \bar{f}(s) ds$	D)	$\frac{1}{s}\bar{f}(s)$
16.	If $L\{f(t)\} = \frac{1}{s^2}$, then $L\left\{\frac{f(t)}{t}\right\}$ is -----			
	A)	$\frac{-2}{s^3}$	B)	$\frac{2}{s^3}$
	C)	$\frac{1}{s}$	D)	$\frac{-1}{s}$
17.	$L\{e^{-2t}\} =$ -----			

	A)	$\frac{1}{s+2}$	B)	$\frac{1}{s-2}$
	C)	$\frac{s}{s+2}$	D)	$\frac{s}{s-2}$
18.	If $L\{f(t)\} = \bar{f}(s)$, then $L\{e^{-2t}f(t)\}$ is -----			
	A)	$\bar{f}(s+2)$	B)	$\bar{f}(s-2)$
	C)	$\bar{f}(s)$	D)	$\frac{1}{2}\bar{f}(s)$
19.	$L\{t \sin t\} =$ -----			
	A)	$\frac{2s}{(s^2-1)^2}$	B)	$\frac{2}{(s^2+1)^2}$
	C)	$\frac{(2s)}{(s^2+1)}$	D)	$\frac{(2s)}{(s^2+1)^2}$
20.	$L\{t^6\} =$ -----			
	A)	$\frac{5!}{s^5}$	B)	$\frac{5!}{s^6}$
	C)	$\frac{6!}{s^7}$	D)	$\frac{6!}{s^6}$